

The Fidelity–Utility Paradox in Surrogate-Based Reinforcement Learning: A Role-Separated Hybrid Methodology in an HVAC Control Case Study

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Abstract

Deep reinforcement-learning (RL) controllers for HVAC are often trained against fast neural-network surrogates, under the assumption that a more accurate surrogate provides a better training environment. We test this assumption on the BOPTEST `bestest_air` testcase and report a negative result, the *fidelity–utility paradox*: the surrogate with lower predictive error can be the worse environment for policy-gradient search. A grey-box RC surrogate with a neural residual heat-flow head (GB) attains a 24-hour rollout RMSE of 0.644 °C versus 1.557 °C for a black-box surrogate (BB), yet a policy trained directly on GB collapses on every seed: its BOPTEST maintenance score is $m_s = 1.07/1.13$ peak/typical (lower is better; $m_s > 1$ is unusable, $N=3$), with comfort violation $> 75\%$, whereas the weaker BB trains a usable controller ($m_s = 0.073/0.095$). Retraining BB at finer resolution makes it *more* accurate yet unusable ($m_s > 1.1$), showing that the driver is temporal resolution and the coarse per-step increment, not model class alone. We then formulate a role-separated hybrid methodology: BB supplies smooth rollout dynamics, while a frozen GB acts only as a per-step model-disagreement reward censor. This role separation gives the best cross-window robustness ($m_s = 0.041$, sub-5% violation) at $\sim 85\times$ live-simulator throughput, with a controller-family-specific censor weight. Across three hydronic transfer cases, recalibration transfers but the frozen controller does not, indicating a component-level rather than universal transfer boundary.

Keywords: reinforcement learning; HVAC control; surrogate model; physics-informed calibration; transfer learning; BOPTEST

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Nomenclature

Symbol	Unit	Meaning
$T_{\text{zone}}, T_{\text{amb}}, T_{\text{sup}}$	°C	zone / ambient / supply-air temperature
C_{zon}	J/K	zone thermal capacitance
a_t	–	control action at step t
$m_s = r_{\text{time}} + r_{\text{sev}}$	–	live BOPTEST maintenance score (time-in-violation + severity; lower better)
RMSE_T	°C	live closed-loop zone-temperature RMSE
Violation	%	fraction of steps outside the comfort band
λ_T, λ_P	–	hybrid temperature / power censor weights
τ	–	transfer pass threshold $1.25 m_{s,\text{PI}}$
ρ_C	–	re-identified C_{zon} ratio vs <code>bestest_air</code>

The complete symbol list (RL formulation, full metrics, transferability, abbreviations) is given in the Supplementary Nomenclature.

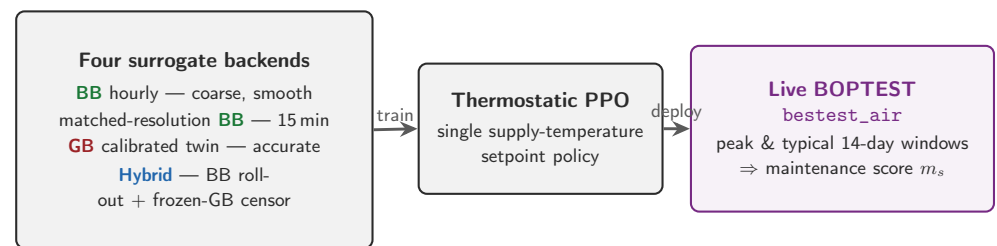
1. Introduction

Buildings account for roughly one third of global final energy use, and heating, ventilation, and air-conditioning (HVAC) systems are their single largest consumer [1,2], making HVAC control a high-leverage decarbonization target. It is also intrinsically hard: zone thermodynamics are nonlinear, coupled, and driven by time-varying occupancy and weather. The proportional–integral (PID) and rule-based controllers used in practice are robust but cannot exploit this structure [3,4], whereas model predictive control can, in principle, but needs an accurate, continuously maintained building model that is expensive to identify and brittle under mismatch [5].

Deep reinforcement learning (DRL) is an attractive alternative because it learns a control policy directly from interaction, without an explicit model, and standardized runtime environments such as BOPTEST now make such controllers reproducibly comparable [1,6]. The obstacle is the sim-to-real deployment gap: policy-gradient algorithms consume millions of environment steps, so training on a live building, or on a high-fidelity physics simulator stepped through an HTTP interface, is prohibitively slow and risks sustained comfort violations during exploration [5,7]. The community’s response is to train the policy against a fast neural-network surrogate of the building, increasingly a physics-informed digital twin (e.g. resistance–capacitance or Neural Ordinary Differential Equation models) that generates state transitions orders of magnitude faster than real time [8,9]. This paradigm proceeds on a rarely-examined assumption, which this study makes explicit and tests: that a surrogate with higher predictive fidelity is automatically a better environment in which to train a controller.

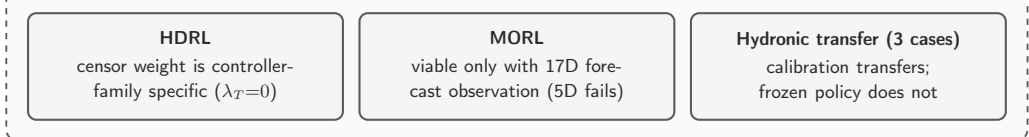
We find this assumption can fail, exposing the *fidelity–utility paradox* (Fig. 1): the surrogate with the lowest multi-step rollout error can be the worst environment for policy-gradient search. Trained directly on the high-fidelity physical twin, the policy converges to a near bang-bang law and fails on the live BOPTEST runtime ($m_s > 1$, comfort violation above 77%), even though that twin is the more accurate offline predictor. The driver is measurable on the surrogate side: the higher-fidelity backends expose a much rougher action→next-temperature response surface, which policy-gradient search exploits into a policy that does not survive distribution shift to the live simulator [10]; Section 4.2 isolates the coarse per-step increment magnitude as the operative property. The methodological point is that predictive fidelity, corpus resolution, and control utility must be separated by controlled decomposition, not compared as a single RMSE number.

1. Primary paradox test (the central causal claim)



Paradox: higher predictive fidelity (calibrated GB, matched-resolution BB) collapses in live control ($m_s > 1$); only the coarse hourly BB and the role-separated hybrid stay usable.

2. Boundary checks (scope of the recipe — not part of the central test)



Live metric: m_s = time-in-violation + worst-case severity ($m_s > 1$ = unusable); also comfort-violation %, energy, and 24 h rollout RMSE_T.

Figure 1. Study overview. *Top, primary paradox test (the central causal claim):* four surrogate backends (coarse hourly BB, matched-resolution BB, calibrated GB, and the role-separated hybrid) each train a thermostatic-PPO controller scored live on BOPTEST `bestest_air`; the higher-fidelity backends (GB, matched-resolution BB) collapse ($m_s > 1$) while the coarse BB and the hybrid stay usable. *Bottom, boundary checks (dashed box, secondary):* HDRL, MORL, and a three-case hydronic transfer study map the scope of the recipe and are *not* part of the central test.

Two secondary gaps accompany the central one. HVAC control is partially observable: a small (5-dimensional) instantaneous observation is too impoverished for forecast-dependent and preference-conditioned control [4,11], and a single physics-regularization strength need not suit thermostatic, hierarchical, and multi-objective agents alike. Existing studies seldom separate surrogate fidelity from controller utility, rarely scrutinize the observation interface, and almost never pre-specify cross-testcase transfer, the methodological gap addressed here.

The proposed method is a hybrid, role-separating architecture that decouples prediction from training (Fig. 1). A compact control-oriented black-box surrogate (BB) supplies smooth, control-friendly Markov rollout dynamics, while an inverse-calibrated physics-informed twin (GB), whose zone thermal capacitance C_{zon} is identified from telemetry, acts as a *frozen per-step reward-shaping censor*. The censor adds a disagreement penalty to the reward, not a term to the policy loss, and the frozen GB never becomes the rollout model, so the policy keeps learning over BB's smooth rollout dynamics while state-action regions on which the physics-informed twin disagrees with the rollout surrogate are discouraged. To address partial observability, the controller interface is widened to a 17-dimensional, forecast-augmented direct-supply-temperature observation. The recipe is evaluated across three controller families: thermostatic proximal policy optimization (PPO), hierarchical DRL (HDRL) [12], and preference-conditioned MORL [11]; the resulting frozen policies are then subjected to a pre-specified transferability study on a hydronic family of BOPTEST testcases.

The evidence hierarchy is deliberately narrow. The primary causal test is the four-backend thermostatic-PPO comparison: coarse BB, matched-resolution BB, direct GB, and the hybrid BB/GB backend. HDRL, MORL, and hydronic transfer are not used to enlarge that central claim; they are boundary tests that identify where the role-separated recipe does and does not transfer. This separation prevents predictive fidelity, training utility,

controller-family regularization, and cross-testcase transfer from being interpreted as one undifferentiated result.

We separate four questions that are often conflated, stated as pre-specified, falsifiable claims (H1–H4) and reported whether supported or falsified; H1–H2 are the primary paradox and its remedy, while H3–H4 are boundary tests that map the recipe’s scope:

- H1. *Fidelity–utility*: a surrogate with higher predictive fidelity is a better RL training environment.
- H2. *Role separation*: using the physical twin as a frozen reward-shaping censor over a smooth black-box rollout recovers the control utility that direct use of the twin destroys.
- H3. *Censor-weight universality*: a single model-disagreement censor weight is sufficient (transfers) across controller families. (Falsification of this hypothesis is what establishes controller-family specificity.)
- H4. *Transferability*: the inverse-calibration pipeline and the frozen controller transfer to related hydronic testcases under documented recalibration.

Stating these in advance lets the negative outcomes be read as findings rather than tuning artefacts: direct-GB collapse, censor-weight non-universality (H3 falsified), and regime-dependent policy transfer. In particular, the calibration pipeline transfers more reliably than the frozen controller, the component-level boundary identified by the study.

The study makes three contributions:

1. **Predictive fidelity is separated from RL training utility.** A calibrated GB twin is the best predictor, but direct PPO training on it collapses in live BOPTEST. A matched-resolution black-box BB ablation shows that the failure is not a grey-box artefact: increasing temporal fidelity can also destroy control utility.
2. **The collapse is diagnosed and repaired by role separation.** The collapsing single-model backends expose rougher action-response surfaces and produce near bang-bang policies. The hybrid backend keeps BB as the smooth rollout model and uses frozen GB only as a reward-shaping censor.
3. **The transfer boundary is component-level.** The calibration pipeline transfers across hydronic testcases, but frozen-policy transfer is regime-dependent. HDRL and MORL further show that the model-disagreement censor weight is controller-family specific.

The remainder of the paper is organized as follows. Section 2 positions the work against HVAC RL, surrogate modelling, and transfer literature; Section 3 gives the experimental taxonomy and workflow; Sections 4.1–4.5 present the three evidence blocks; and Section 5 closes with synthesis, limitations, and conclusions.

2. Related Work

Reproducible HVAC-control research depends on a shared benchmark, because controller claims otherwise vary with the simulator, weather year, disturbance schedule, and comfort definition. BOPTEST serves Modelica building models through a uniform API [6,13] with a Gym interface for RL workflows [14], complementing demand-response benchmarks such as CityLearn [15,16]; every controller here is finally re-scored in BOPTEST, not only in its learned environment. Against this benchmark, HVAC DRL studies compare learned policies with rule-based, Guideline 36, or MPC references [2,3,7,17] and extend them to multi-agent and context-aware settings [18,19]; hierarchical policies decompose comfort/energy objectives across timescales [12,20,21], and MORL represents the energy–comfort trade-off explicitly [11,22–24]. These works fix the training environment, which is precisely what this study varies.

Learned building surrogates make DRL-scale training feasible where direct simulation is too slow, spanning black-box networks, grey-box RC models, Neural ODEs, and physics-informed learning [25–33], and connecting to the broader scientific-machine-learning surrogate literature on operator learning and projection-based reduced-order models [34–36]; Hou and Evins [8] frame surrogate development as a staged validation protocol and fast-surrogate work emphasises runtime feasibility [9]. Most of this literature validates prediction metrics, yet a controller (like a surrogate-based optimizer that can drive an accurate surrogate into infeasible optima by exploiting its residual error [37]) uses the surrogate as a closed-loop transition function under policy-induced states, so we report predictive fidelity and runtime first, then test the stronger claim that fidelity implies training utility.

That stronger claim is a distribution-shift question. On-policy data collection and observation design shape the states an RL controller learns on [4,38,39], and safe-RL work shows that small model errors become operationally important once actions are optimised against them [7,40]. This links HVAC control to model exploitation and dataset shift in model-based RL [10,41], and specifically to the *objective mismatch* between a surrogate trained for predictive likelihood and one that yields a good downstream policy [42,43]; decision- and value-aware model learning answer this mismatch by retraining the model toward the control objective [44], and the same low-error-yet-poor-transfer effect is central to sim-to-real deployment [45]. A low test-set RMSE thus does not guarantee correct behaviour on policy-induced states, and our negative direct-GB result is a control-oriented instance of this mismatch. Unlike value-aware retraining, we hold both surrogates fixed and instead *measure* the mismatch mechanism, action-response roughness induced by temporal resolution, and resolve it by *role separation* rather than by changing the surrogate’s training objective.

Finally, transfer-oriented HVAC work reduces the cost of adapting DRL controllers across buildings, zones, and climates [5,46–48], and world-model/Dyna control motivates learning in an internal model before deployment [49,50]. These lines rarely separate two questions we keep distinct: whether the surrogate recalibrates to the target physics, and whether a frozen controller stays deployable through a new actuator interface, which our version-locked hydronic transfer study answers separately.

3. Experimental Design

This section defines the experimental objects used throughout the paper: the BOPTEST control problem, the surrogate taxonomy, the hybrid backend, the controller families, and the evaluation metrics. The detailed numerical instantiation of every quantity defined here (network sizes, calibrated parameters, hyperparameters, and measured scores) is reported with the corresponding evidence in Section 4; this section fixes the notation and method definitions before the experimental evidence is reported.

Table 1. Experimental taxonomy used throughout the paper. The first four backends define the central fidelity–utility test; HDRL, MORL, and hydronic transfer are boundary tests.

Label	Step/corpus	Architecture	Role in paper
BB hourly	3600 s	black-box	Smooth rollout baseline and positive control in the predictive-validation and central-control experiments.
matched-resolution BB	900 s	same black-box	Temporal-resolution ablation used to isolate whether BB’s utility comes from smoothing rather than black-box structure.
raw GB	900 s	grey-box RC	Uncalibrated predictive reference used only to quantify the Stage A/B/C calibration gain.
calibrated GB	900 s	grey-box RC + Stage A/B/C	High-fidelity physical twin and direct-training negative control in the central fidelity–utility test.
hybrid	BB rollout + frozen GB	role-separated backend	Remedy: BB supplies smooth rollout dynamics, while frozen GB acts only as a model-disagreement reward censor.

Only BB hourly, matched-resolution BB, calibrated GB, and the hybrid backend are part of the central fidelity–utility test. HDRL, MORL, and hydronic transfer are reported later as boundary and transferability tests rather than as additional evidence for a universal controller claim. Detailed algorithmic objectives are provided in the Supplementary Material; the main text retains the equations required to interpret the reported metrics, control interface, hybrid censor, and transfer threshold.

3.1. BOPTEST task and evaluation metric

The plant is a single-zone building emulator with heating and cooling actuation, drawn from the BOPTEST family of Modelica-based test cases [6,13] and served to the controller through BOPTEST’s uniform HTTP interface [14]. Formally the emulator is a continuous-time dynamical system whose latent thermal state evolves under a scalar control action $a_t \in [-1, 1]$ (mapped to a supply-temperature set-point by Eq. (2)) and an exogenous disturbance stream d_t (ambient weather, solar gain, occupancy, and time-varying comfort set-points). The environment is queried at a fixed control period $\Delta t = 900$ s (15 minutes), yielding the discrete-time transition $s_{t+1} = \Phi(s_t, a_t, d_t)$ that all surrogates are trained to approximate. The primary test case is `bestest_air`; a hydronic family of related test cases is held out for the transferability study (Section 4.5).

Primary test case.

`bestest_air` is a single-zone office of $6 \times 8 \times 2.7$ m (48 m^2) whose envelope follows the ASHRAE Standard 140 BESTEST Case 900 definition (heavyweight concrete-block walls, an insulated slab, and 12 m^2 of south glazing). Conditioning is an idealized four-pipe fan-coil unit: a gas boiler (constant efficiency 0.9) feeds the heating coil and a chiller (constant COP 3.0) the cooling coil, with a variable-speed supply fan. The zone is a two-person office occupied 08:00–18:00 with light internal gains and occupied comfort set-points $21/24^\circ\text{C}$, under Denver-Stapleton (CO, USA) TMY weather. Of the available BOPTEST actuators, the controller commands only the supply-air-temperature set-point, which the action map (2) restricts to $18\text{--}35^\circ\text{C}$, and observes zone temperature, CO_2 , and power.

Transfer family.

The transfer set contains three single-zone-like hydronic BOPTEST cases with different heat sources, distribution paths, and actuator interfaces (Table 2); they form an

increasing-difficulty ladder away from `bestest_air`, share the single-zone first-order energy balance (6), and are bridged by the per-test-case adapters of Section 4.5 (per-case schematics in Supplementary Figure S29).

Table 2. BOPTEST test cases used in this study, from the official BOPTEST documentation [6]. All are single-zone or single-zone-like and heating-capable.

Test case	Type	Floor area	Heat source / emitter	Climate
<code>bestest_air</code> (source)	Office	48 m ²	Gas boiler + chiller / 4-pipe fan-coil	Denver
<code>bestest_hydronic</code>	Residential	48 m ²	Gas water heater / radiator (5 kW)	Brussels
<code>bestest_hydronic_heat_pump</code>	Residential	192 m ²	Air-to-water heat pump (15 kW) / floor loop	Brussels
<code>singlezone_commercial_hydronic</code>	Commercial	8500 m ²	District heating / AHU + radiator circuit	Copenhagen

Problem formulation.

Because the controller never observes the full latent state, control is a partially observable Markov decision process (POMDP) $\langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \Phi, R, \gamma \rangle$, where the agent receives an observation $o_t = h(s_t, d_t) \in \mathcal{O}$ and selects $a_t \sim \pi(\cdot | o_t)$ so as to maximise the expected discounted return [51]

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right], \quad r_t = -(w_c c_t + w_e e_t), \quad (1)$$

with c_t a thermal-discomfort term, e_t the heating energy over the step, and w_c, w_e trade-off weights. This scalarized comfort–energy objective follows the standard multi-objective RL treatment of HVAC control, where preferences are represented by explicit weights rather than hidden inside a single black-box reward [22–24]. The comfort component is piecewise-linear with an asymmetric, ambient-dependent slope (steeper penalties for cold-weather undershoot, fixed a priori from this physical asymmetry rather than tuned to any backend), plus a small in-band bonus, and the energy component is proportional to heating power; the full functional form and its fixed parameters are given in the Supplementary Material. Because HVAC control is partially observable, a five-dimensional instantaneous observation is insufficient and a forecast-augmented one is required [4]: the canonical interface is a 17-dimensional observation (physical states, cyclic time features, ambient forecasts, and previous actions; full per-feature breakdown in Supplementary Table S12), whereas the failed MORL baseline uses only the earlier 5D observation. The agent acts through a single normalized supply-temperature command,

$$a_t \in [-1, 1], \quad T_{\text{sup},t} = 18 + \frac{a_t+1}{2}(35-18) \quad [^{\circ}\text{C}], \quad (2)$$

with a 1.0 °C deadband, a per-step rate limit, and a 21–24 °C comfort band. The controller’s action space is one-dimensional throughout: the policy emits only this scalar supply-temperature command. The surrogate transition is *trained* on a two-dimensional command (a_0, a_1) (the supply-temperature action a_t of Eq. (2) together with a normalized fan setting a_1 , both excited only while the offline corpus is generated), but during every RL rollout and every live BOPTEST evaluation the fan is held at its BOPTEST baseline (a_1 fixed) and only the supply-temperature set-point is commanded, which is also the single input that BOPTEST overwrites (Section 3.4). For *evaluation* (as opposed to training) deployed controllers are scored with the duration-and-severity maintenance score

$$m_s = r_{\text{time}} + r_{\text{sev}}, \quad (3)$$

the time-in-violation plus worst-case relative-severity metric of Wang et al. [7]; lower is better and $m_s > 1$ marks an unacceptable controller. Explicitly, $r_{\text{time}} = N_{\text{viol}}/N \in [0, 1]$ is the fraction of control steps in which the zone temperature leaves the comfort band $[T_{\text{lo}}, T_{\text{hi}}]$, and $r_{\text{sev}} = \max_t \max((T_{\text{zone},t} - T_{\text{hi}})/T_{\text{hi}}, (T_{\text{lo}} - T_{\text{zone},t})/T_{\text{lo}}, 0)$ is the largest single-step band exceedance normalised by the band bound; both terms are dimensionless. Because $r_{\text{time}} \leq 1$, the threshold $m_s > 1$ flags a controller that is either out of band for a large fraction of the horizon or commits a severe worst-case excursion, the regime treated here as a failed deployment. Because r_{sev} normalises the excursion by the Celsius band bound, the $m_s > 1$ cut is a reporting convention inherited from Wang et al. [7] rather than a physically invariant boundary. This separation of a shaped training reward (1) from an unshaped safety score (3) follows safe-RL evaluation practice [40] and the HVAC safety-score construction used by Wang et al. [7]. Thus m_s is not introduced as a new physical quantity; it is an operational safety metric adapted from prior HVAC DRL evaluation work and implemented consistently across all BOPTEST runs.

3.2. Surrogate backends and the paradox test

All three surrogates are fast approximations of Φ used as the training environment in place of the slow emulator, in the model-based / Dyna tradition of learning a model and planning or training a policy against it [49,50]. Each is fit to BOPTEST-generated transition tuples (s_t, a_t, d_t, s_{t+1}) and used as a *deterministic point-transition* map: during every RL rollout the next state is set to $\hat{s}_{t+1} = f_{\theta}(s_t, a_t, d_t)$ (or its GB analogue g_{ϕ}), not sampled from a learned conditional distribution $P_{\theta}(\cdot | s_t, a_t, d_t)$. This is appropriate where the zone's conditional dynamics are effectively deterministic given the observed drivers; it does not represent transition uncertainty, a recognised limitation for the noisier power channel (Section 5). The full field-by-field transition-dataset schema (tagging each field as a BOPTEST measurement, an exogenous disturbance, a derived feature, a control action, or a regression target, and giving the recursive-rollout state update) is provided in the Supplementary Material. They differ in how much building physics is hard-wired into their structure.

Control-oriented surrogate BB.

BB is a compact black-box neural one-step predictor $\hat{s}_{t+1} = f_{\theta}(s_t, a_t, d_t)$, trained by supervised minimisation of a multi-step rollout error so that long-horizon trajectories, not just one-step residuals, stay accurate. Its design goal is not maximal predictive accuracy but a *smooth, well-conditioned* response surface for policy-gradient search; the architecture and parameter count are reported in Section 4.1.

Physically informed surrogate GB and Stage A/B/C calibration.

GB replaces the black box with a grey-box resistance–capacitance (RC) thermal network whose unmodelled heat flow is supplied by a neural residual head. It is an ODE-structured grey-box surrogate, a continuous-time RC balance, in the spirit of Neural ODEs [30] and classical RC building identification [25–28], but integrated discretely by an explicit Euler step at the control period Δt rather than by an adaptive ODE solver. Writing the zone capacitance as C_{zon} and a lumped envelope resistance as R , the core balance is

$$C_{\text{zon}} \frac{dT}{dt} = \frac{T_{\text{amb}} - T}{R} + \eta a_t \dot{Q}_{\text{max}} + q_d,$$

with $\eta a_t \dot{Q}_{\text{max}}$ the delivered heating power and q_d aggregated disturbance gains (the neural residual head). Embedding this conservation law in the network makes it physics-informed in the sense of Raissi et al. [29], Karpadne et al. [31], Willard et al. [32], so its

parameters are physically interpretable; the analysis nonetheless treats C_{zon} as an *effective* thermal capacitance under the chosen surrogate structure, a lumped parameter that may absorb unmodelled fast dynamics, rather than a direct measurement of zone air capacitance. The parameters are recovered by a staged inverse calibration (Stages A/B/C) in the spirit of staged grey-box calibration [25], and the identifiability of C_{zon} is checked with a Fisher/Laplace confidence interval; the staged protocol and the resulting calibrated values are detailed in Section 4.1.

3.3. Role-separated hybrid remedy

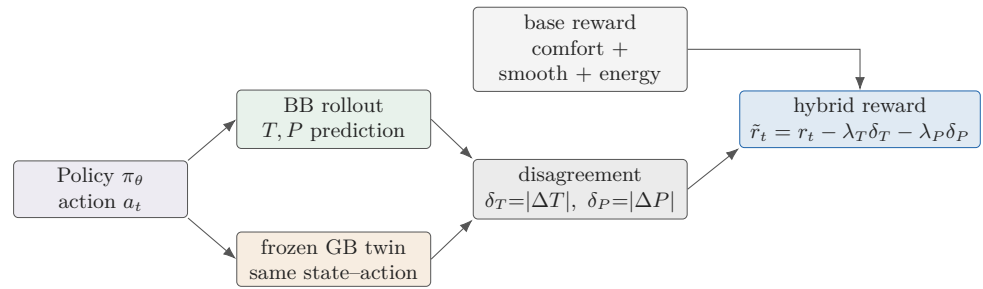
The hybrid backend is not introduced as a new reinforcement-learning algorithm. It is an application-specific role-separation strategy grounded in three existing lines of work: model-based/Dyna and world-model training in learned environments [49,50], physics-informed and grey-box surrogate modelling [25,29,31,32], and safe-RL reward shaping or constraint penalties [7,40]. The contribution here is the allocation of roles inside surrogate-trained HVAC control: BB supplies the smooth rollout dynamics, while a *frozen* GB acts only as a per-step reward-shaping sensor. Training uses the shaped reward

$$\tilde{r}_t = r_t - \lambda_T \delta_{T,t} - \lambda_P \delta_{P,t}, \quad (4)$$

where the disagreement terms are the per-step L_1 (absolute) distances between the one-step predictions of the rollout surrogate f_θ (BB) and the frozen twin g_ϕ (GB), separately for the zone temperature and the HVAC power channel,

$$\delta_{T,t} = |f_{\theta,T}(s_t, a_t, d_t) - g_{\phi,T}(s_t, a_t, d_t)| [^\circ\text{C}], \quad \delta_{P,t} = |f_{\theta,P}(s_t, a_t, d_t) - g_{\phi,P}(s_t, a_t, d_t)| [\text{W}],$$

with no squaring, normalisation, or clipping. The two sensor weights differ in scale because they multiply a temperature residual (in $^\circ\text{C}$) and a power residual (in W); for the thermostatic controller $\lambda_T = 0.10$ and $\lambda_P = 5 \times 10^{-5}$. Crucially, g_ϕ is evaluated *forward only* and its outputs are detached, so the sensor enters the *reward* alone, never the policy loss, and never the rollout dynamics, which remain those of BB. PPO estimates its policy gradient from sampled returns rather than by differentiating the environment, so this role split does not preserve any environment-to-policy gradient; what it preserves is the smooth BB *trajectory-and-return distribution* of Section 3.2 (and its favourable optimisation conditioning), while penalising state-action regions on which the frozen twin disagrees with the rollout. In other words, GB does not replace BB, form an accuracy-oriented ensemble, or introduce a second policy-loss term. The penalty measures disagreement between two imperfect surrogates, not violation of a conservation law or a physically feasible set; it is therefore a *model-disagreement* sensor (physics-informed through GB's structure, but not a hard physical-feasibility filter), applied forward-only over the smoother BB training environment (Figure 2). The swept values of λ_T per controller family are given in Section 4.2.



The frozen GB is a forward-only reward-shaping censor
— it is not a policy-loss term and not the rollout model.

Figure 2. Hybrid reward-shaping mechanism. The black-box surrogate (BB) supplies the smooth rollout dynamics; the frozen grey-box surrogate (GB) is evaluated forward-only on the same state-action pair and enters *only* as a per-step reward penalty on BB–GB disagreement (4). GB is neither the rollout model nor a policy-loss term.

Controller families.

All policies are optimised with proximal policy optimization (PPO) [52] using generalized advantage estimation [53] and the Stable-Baselines3 implementation [54]; PPO maximises the standard clipped surrogate objective with generalized-advantage estimates (full form in the Supplementary Material). The shared PPO settings are learning rate 3×10^{-4} , discount factor $\gamma = 0.99$ (an effective horizon of ~ 25 h at the 900 s step), and ten optimization epochs per rollout; the families differ in rollout length, minibatch size, and total timestep budget. The surrogate recipe is evaluated across three controller families of increasing structure: (i) a *thermostatic* PPO agent on the shared forecast-augmented observation; (ii) a *hierarchical* DRL controller (HDRL) that decomposes control across timescales, in the options/feudal tradition [20,21] as recently applied to year-round HVAC operation [12]; and (iii) a *preference-conditioned* multi-objective controller (MORL) that conditions on a comfort–energy preference vector [11,22–24] and acts on the forecast-augmented observation. All families share the 17D forecast-augmented observation and supply-temperature action (2) of Section 3.1; the failed MORL baseline instead uses a 5D instantaneous observation.

3.4. Evaluation and transfer protocol

Control is applied to the live plant through the BOPTEST Run-Time Environment (RTE): a Dockerised Modelica emulator with embedded baseline (local-loop and supervisory) controllers, a disturbance forecaster, and a KPI calculator, all exposed through a RESTful HTTP API [6]. The agent does not replace the plant; it *overwrites* one control input, the supply-temperature set-point, while the remaining embedded loops run at baseline. At each 900 s control step it reads measurements y_t (zone temperature, CO₂, power) and disturbance forecasts ω_t , forms the 17D observation, maps its action to a supply temperature through (2), writes that set-point through the BOPTEST overwrite pair, and advances the emulator one step; the maintenance score and other KPIs are then computed by the RTE from the resulting live trajectory rather than from the surrogate. The hydronic transfer cases reuse the same supply-temperature command, routed through the per-test-case adapters of Section 4.5. Figure S25 details the loop and the overwrite-signal mechanism.

Surrogate predictive fidelity is reported with the ASHRAE-style calibration statistics CV(RMSE) and NMBE over multi-step rollouts, the standard accuracy measures for calibrated building models [25]. Downstream control utility is reported with the maintenance score (3) [7], the comfort-violation fraction, and heating energy, each benchmarked against BOPTEST’s built-in baseline controller. Because a policy trained on a surrogate is deployed

on a different (live) system, the central methodological risk is dataset/distribution shift [41]; the evaluation therefore always reports the *live*-emulator score, never only the surrogate-internal score. The transfer-gap diagnostics used later (Δm_s , action-gap norm, and first divergence step) operationalize this same simulator-induced distribution-shift idea for the surrogate-to-BOPTEST setting [10,41]; they are diagnostic adaptations, not universal benchmark metrics. For transferability, each hypothesis (H1–H4), pass threshold, and recalibration adapter is pre-specified *before* the corresponding runs; full reproducibility details are in the Supplementary Material. Cross-test-case transfer is realised through a light recalibration adapter rather than assumed zero-shot, following the evidence that transfer is source-dependent [5]. Controller scores are reported with seed statistics wherever repeated-seed runs are available: the central fidelity–utility backends (pure BB, matched-resolution BB, direct GB, and the hybrid) are evaluated over $N = 3$ seeds (seed s.d. shown where available; Fig. 5, Table S14) and MORL over the five fixed seeds $\{42, 43, 44, 45, 46\}$, whereas the HDRL censor-weight sweep remains a deterministic single-seed run, a limitation stated explicitly in Section 5. The statistical protocol is summarized here; the corresponding results are reported in Sections 4.1–4.5.

Implementation.

All controllers are evaluated on BOPTEST [6,13,14], using `bestest_air` as the primary testcase and a `bestest_hydraulic` family for transfer (Section 4.5), at a fixed 15 min control period through one HTTP contract behind which four interchangeable backends sit (live BOPTEST, BB, GB, hybrid), so only the dynamics backend changes across the surrogate-versus-live comparison, and the surrogate backends run $85.0\text{--}114.2\times$ faster than the live loop (a CPU-measured throughput benchmark), making policy-gradient training tractable on commodity hardware. Surrogates use three corpora generated by scripted excitation: a canonical BB corpus (51,200 at 1 h), a 15-minute exploration corpus (48,384 at 900 s), and a GB calibration corpus (10,744 at 900 s); the deliberate one-hour/fifteen-minute pairing is the instrument behind the matched-corpus ablation. BB (8,482 parameters) is fit by multi-step rollout regression; GB is identified by the Stage A/B/C inverse calibration of Section 3.2 [25], yielding $C_{\text{zon}} = 4.413 \times 10^5 \text{ J/K}$ (+5.1% over the physical prior). All controllers are trained with PPO [52] in Stable-Baselines3 2.1.0 [54] under Python 3.11.9; the hybrid loads the frozen GB as the censor of Eq. (4). All experiments ran on a laptop workstation (AMD Ryzen 7 7445HS, 6 cores/12 threads, 3.2 GHz base; 32 GB RAM; NVIDIA RTX 4050 Laptop GPU) under Windows 11 Pro, with the BOPTEST RTE deployed as a Docker container; the surrogate-versus-live speedup was timed on CPU. Canonical targeted-window KPI tables use seed 42; the central BB, matched-resolution BB, direct-GB, and hybrid thermostatic comparison additionally carries an $N = 3$ seed-robustness audit (Table S14); the central paradox claim thus receives seed replication while the boundary tests receive effort proportional to their role (the HDRL mechanistic control remains single-seed and MORL is evaluated over $N = 5$ fixed seeds with stated variance); all are benchmarked against BOPTEST’s built-in PI [7]. Each hypothesis (H1–H4) is pre-specified *before* its runs; full reproduction details are in the Supplementary Material.

4. Experimental Results

The experiments answer four questions in sequence. *First*, which surrogate is the more faithful predictor? The calibrated GB RC–NeuralODE, by a wide margin (§4.1). *Second*, does that higher fidelity make it the better reinforcement-learning training environment? No: used directly, both GB and a matched-resolution BB collapse on the live runtime (§4.2), which is the paper’s central causal test. *Third*, why do the high-fidelity backends collapse, and how is control recovered? Their action-response surfaces are rougher, which drives

policy saturation and live collapse, whereas a role-separated hybrid (smooth BB rollouts plus a frozen GB reward censor) restores it (§4.3). *Fourth*, how far does the result transfer? It is controller-family specific (§4.4), and the inverse-calibration pipeline transfers across buildings while the frozen policy does not (§4.5). These last two subsections are *boundary tests* that map the scope of the paradox rather than re-proving it. The corresponding reproducibility details are retained in the Supplementary Material.

4.1. Predictive fidelity: establishing the asymmetry

The calibrated GB RC–NeuralODE is the more faithful predictor, reaching a 24-hour rollout RMSE of 0.644 °C against 1.557 °C for the compact BB. This predictive-validation experiment establishes that digital-twin result by contrasting the two surrogates across architecture, calibration, data resolution, rollout accuracy, and runtime; whether the more accurate twin is the better PPO training environment is tested in the central fidelity–utility experiment (Section 4.2).

This subsection asks whether the surrogate family is credible as a fast, physically interpretable digital twin. It does not yet ask whether a controller trained on that surrogate transfers to the live BOPTEST RTE. The fixed experimental order is:

1. collect and train the BB direct- T_{sup} surrogate;
2. prepare the 15-minute corpus and run GB Stage A/B/C inverse calibration;
3. retrain BB on the same 15-minute corpus to bound the calibration claim;
4. build the Hou-and-Evins reporting tables and the speed benchmark.

4.1.1. Surrogate construction and inverse calibration

All predictive-validation surrogates share a single *direct supply-temperature* (direct- T_{sup}) parameterization: the policy commands the supply-air setpoint directly (the two-dimensional action $a = (a_0, a_1) \in [-1, 1]^2$ maps affinely to the supply temperature through Eq. (2) and to a fan command), and every surrogate exposes the same transition signature $(T_{\text{zone}}, T_{\text{amb}}, h, d, a_0, a_1) \mapsto (\hat{T}_{t+1}, \hat{P}_{t+1})$ with $\hat{P} \geq 0$ (Softplus output, capped at $P_{\text{max}} = 5500$ W). A single adapter wraps four interchangeable backends behind this interface (legacy BB, raw GB, calibrated GB, and the hybrid pairing), which is what makes the like-for-like fidelity comparisons in this section well posed: only the internal dynamics change, never the control interface or the observation map (interface details and modelling assumptions in Supplementary Table S31).

The BB surrogate is a compact, control-oriented model: its role is not long-horizon accuracy but smooth, fast rollout dynamics for downstream PPO. It predicts a temperature increment and a power head, $\hat{T}_{t+1} = T_t + f_{\theta,T}(x_t)$ and $\hat{P}_{t+1} = f_{\theta,P}(x_t)$, trained by a multi-horizon (1-, 2-, 4-step) supervised rollout objective with a physical-consistency penalty (full form in the Supplementary Material); the update carries *no* explicit time step, a design choice revisited in Section 4.1.3. The canonical checkpoint is the early-stopped optimum of a 215-epoch run (best validation epoch 185, held-out one-step $R^2 = 0.991$), a single-step score deliberately optimistic relative to the multi-step rollout error (Section 4.1.2) that actually governs whether the surrogate can substitute for BOPTEST. Its input vector is deliberately low-dimensional (thermal state, outdoor disturbance, cyclic time embedding, and the direct- T_{sup} action [17]), with a non-negative power head, making BB a control-oriented surrogate rather than a generic forecaster (hyperparameters in Supplementary Table S3).

Physics-informed GB and Stage A/B/C calibration.

Unlike the purely black-box BB head, the GB surrogate is built on a first-principles lumped-capacity zone balance with a single effective parameter, the zone thermal capacitance C_{zon} ,

$$C_{\text{zon}} \frac{dT_{\text{zone}}}{dt} = \dot{Q}(x; \phi), \quad \dot{Q}(x; \phi) = q_{\text{scale}} q_{\phi}(x), \quad (5)$$

where $\dot{Q}$ is a learned net heat-flow head ($q_{\text{scale}} = 3000$ W) that absorbs HVAC input, envelope coupling, and internal gains through the feature vector x ; embedding this thermodynamic structure follows the physics-informed reinforcement-learning paradigm [33]. The balance is integrated by a clipped explicit Euler step, and C_{zon} is a positively reparameterized scalar bounded below by 5×10^4 J/K (integration and reparameterization details in the Supplementary Material). Because C_{zon} is a property of the building rather than of the data-generating policy, it is expected to remain meaningful across testcases, which is what makes the inverse identification below worthwhile.

The canonical GB is calibrated by a staged inverse identification (Supplementary Table S33): Stage A aligns telemetry and removes latency/bias; Stage B identifies C_{zon} as a maximum-a-posteriori estimate under a weak Gaussian prior, restricting the fit to excitation-rich transients where the sensitivity $\partial \hat{T} / \partial C$ is large (quasi-steady windows are uninformative and would merely pull the estimate back to the prior; Table S6); and Stage C refines the neural residual heads with C_{zon} frozen.

The identified $C_{\text{zon}} = 4.413 \times 10^5$ J/K is a +5.1% update over the physical prior, within one standard error (Laplace/Fisher interval in Supplementary Table S33); physically it corresponds to ~ 439 kg of air (~ 366 m³), the expected order for the `bestest_air` zone. As a direct check, sweeping the supply-temperature command with the state held fixed yields the physically correct sign in 100% of 400 sampled states (mean sensitivity +0.061 °C per °C; the full monotonicity sweep is in Supplementary Figure S2).

4.1.2. Predictive fidelity and matched-corpus attribution

For horizon h , predictive accuracy is reported as the multi-step rollout RMSE $\text{RMSE}_T(h) = \sqrt{N^{-1} \sum_i e_i(h)^2}$ of the per-step error $e_i(h) = \hat{T}_i(t+h) - T_i(t+h)$, the engineering-tolerance CDF $F_{|e|}(\epsilon) = \Pr(|\hat{T} - T_{\text{BOTEST}}| \leq \epsilon)$, and its 95th-percentile absolute-error tail P95. The coefficient of determination follows the standard definition $R^2 = 1 - \text{SS}_{\text{res}} / \text{SS}_{\text{tot}}$ against the held-out mean, so a *negative* 24 h R^2 (as for the hourly BB in Table S8) means the long-horizon rollout is worse than predicting that mean.

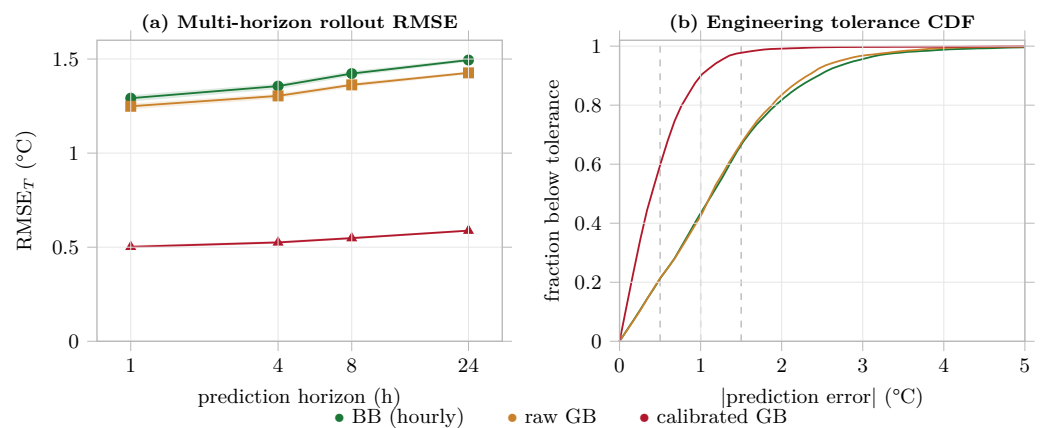


Figure 3. Multi-horizon predictive validity, residual spread, and engineering error CDFs from BOPTEST rollouts.

The calibrated GB surrogate reaches a 24 h rollout $\text{RMSE}_T = 0.644^\circ\text{C}$, compared with raw GB at 1.466°C and the canonical BB hourly reference at 1.557°C (Figure 3).

Table S8 exposes the split between the two families. The hourly BB has a *negative* 24 h R^2 , worse than the historical mean over long horizons, and worse than a naive persistence forecast at 24 h (1.557 vs 0.969°C), yet it remains a usable control surrogate because PPO needs smooth local gradients, not long-horizon realism. Calibrated GB instead keeps a positive 24 h R^2 , bounds its tail (24 h P95 1.207°C), and beats persistence at every horizon, confirming BB is a control surrogate rather than a forecaster.

The episode-level diagnostic is important because a low aggregate RMSE can hide a single unstable rollout. The calibrated GB artifact remains below the 1°C reference in all eight held-out episodes; its best episode is peak heat thermostatic at 0.486°C and its worst episode is typical heat hdlr at 0.964°C . This supports predictive validity as a repeated-episode result rather than a single-window artifact.

The study includes a corpus-matched BB retraining step because the original BB and canonical GB artifacts use different data resolutions. This is not a cosmetic check: without it, a reviewer could correctly attribute the GB gain to the 15-minute corpus rather than Stage A/B/C calibration.

Table 3. Corpus-matched comparison.

Variant	Step (s)	Calibration	24 h RMSE_T
BB hourly	3600	n/a (control-oriented black box)	1.557
BB 15-min matched	900	n/a (control-oriented black box)	0.876
raw GB	900	OFF (no Stage A/B/C)	1.466
calibrated GB	900	FULL (Stage A + B + C, $C_{\text{zon}} = 4.413\text{e5 J/K}$)	0.644

The attribution is an additive split of the legacy-BB-to-calibrated-GB gap into a corpus term (BB hourly $\rightarrow$ BB 15-min) and a calibration term (BB 15-min $\rightarrow$ calibrated GB), shown as the waterfall in Supplementary Figure S27; the split is additive because the intermediate point is BB retrained on the 15-min corpus. Of the total 0.913°C drop from legacy BB to calibrated GB, 0.681°C (74.6%) is the corpus shift and 0.232°C (25.4%) Stage A/B/C (an alternative split through the raw GB backbone gives 9.9%/90.1%; they differ because corpus and architecture interact non-additively). Either way the calibration claim is positive but bounded.

4.1.3. Runtime feasibility and predictive-validation synthesis

The final predictive-validation step builds the speed benchmark. The benchmark compares the same BOPTEST RTE HTTP loop used by the paper against local surrogate stepping under a 900 s control step. The result is not a controller claim; it only establishes feasibility for downstream policy optimization.

At the hybrid throughput, a single 5×10^6 -step policy-optimization run costs about 46.6 minutes of pure environment time, versus roughly 66.1 hours on the live BOPTEST RTE HTTP loop. The calibrated GB surrogate is 114.2x faster than the BOPTEST RTE HTTP loop, and the hybrid reference remains 85.0x faster. This closes the runtime part of the predictive-validation experiment: the surrogate family is not only more accurate after calibration, but also fast enough to support repeated policy-training experiments without using the RTE HTTP loop as the inner training environment.

The canonical BB is trained at a 3600 s step but applied at 900 s; since its transition carries no explicit timestep scaling, the hourly increment is used unchanged, the structural reason the legacy BB carries a 1.557°C 24 h error while the corpus-matched BB reaches 0.876°C (Table 3). The mismatch is kept deliberately: every controller-facing KPI is

measured on live BOPTEST and swapping checkpoints would break the frozen evaluation protocol. The matched-corpus model is evaluated downstream as a closed-loop ablation in the central fidelity–utility test (Section 4.2). Unlike BB, the GB forward pass (5) carries an explicit Δt , so a warm-started GB rescales the inherited BB heat term to keep the per-step increment dimensionally consistent at the 900 s runtime step (scaling factor in the Supplementary Material); this time-scaling is exactly what the BB update lacks.

The predictive-validation experiment closes the predictive-fidelity claim only. Calibrated GB is the best 24-hour predictor, and matched-resolution BB shows that much of the predictive gain comes from temporal resolution. Whether this additional fidelity improves PPO training is a separate control question tested in the central fidelity–utility experiment.

4.2. Higher fidelity destroys RL utility

Higher predictive fidelity does not make a better RL training environment. Used directly as PPO rollout environments, both the more accurate GB and a matched-resolution BB collapse on the live runtime ($m_s > 1.0$), whereas the less accurate hourly BB trains a usable controller. This is the central causal test of the paper, built on the asymmetric premise that BB is the weaker predictor while the calibrated GB RC–NeuralODE is the stronger one. It compares four training backends under the same thermostatic PPO controller: coarse BB, matched-resolution BB, direct GB, and the hybrid BB/GB backend. The aim is not to compare all possible HVAC controllers, but to test whether higher surrogate predictive fidelity improves downstream live BOPTEST utility.

All policies in this central test are trained on surrogate backends and then evaluated in closed loop against the live BOPTEST `bestest_air` runtime. Two targeted 14-day windows are used: `peak_heat_window` (January, daily-mean ambient -24.4°C) and `typical_heat_window` (February, $+2.4^\circ\text{C}$). HDRL, MORL, and the yearly PI reference are reported in Section 4.4 as boundary tests: they identify whether the hybrid regularization and observation design transfer across controller families, not as additional proof of the primary paradox.

Figure S4 summarizes the pipeline. Direct GB is a negative control, not a failed implementation: it asks whether the highest-fidelity predictive surrogate can be used directly as the policy rollout environment (it cannot). The hybrid backend then asks whether the same physical model becomes useful when its role changes from dynamics provider to reward-shaping censor: yes for thermostatic PPO, no for HDRL at the same λ_T , and conditionally yes for MORL once the observation interface is widened from 5D to 17D.

This pipeline is only feasible because the surrogate backends are orders of magnitude faster than the live emulator (Fig. S5): millions of policy-gradient steps that would take days against BOPTEST complete in hours in-process.

The controller-learning experiment is structured as a sequence of falsifiable controller-learning hypotheses (CL1–CL5), each resolved against live BOPTEST evidence and tabulated in Supplementary Table S30. These are distinct from the four paper-level hypotheses (H1–H4) of the Introduction: CL1–CL3 are the controller-experiment counterparts of H1–H3, while CL4–CL5 are additional MORL-specific checks; the paper-level transferability hypothesis H4 is closed separately in Section 4.5. This pre-specification-style framing is what lets the negative results (direct GB, HDRL over-regularization, MORL seed variance) be read as decisions rather than tuning accidents.

The shared 17D observation, supply-temperature action (2), comfort reward, and maintenance score (3) are defined in Section 3.1; RMSE_T is reported against the band centre $T^* = 22.5^\circ\text{C}$. The baseline is a single-level thermostatic PPO trained purely on the BB rollout surrogate (1-hour step, $\lambda_T = \lambda_P = 0$; SB3 MlpPolicy, 10M surrogate steps, Table S11) and run *zero-shot* on the live BOPTEST runtime, the reference against which the

two negative controls (direct GB, warm-start) and the hybrid are measured. In the hybrid, frozen GB acts only as a per-step reward censor (4): across the hybrid traces the mean temperature disagreement is 0.969 °C (p95 2.52) and the mean power disagreement 708 W, large enough to shape learning, bounded enough to stay informative.

Thermostatic PPO negative control and hybrid recovery.

The paradox is summarized as a single measured evidence chain in Figure 4; Table 4 and Figure S7 give the underlying numbers. Pure BB PPO is already usable ($m_s = 0.073$ peak, 0.095 typical). Direct GB PPO fails catastrophically despite GB’s superior predictive-validation fidelity: live violation reaches 77.1% on peak and 82.4% on typical, with RMSE above 4.3 °C. The hybrid backend resolves the conflict: $m_s = 0.087$ on peak and 0.041 on typical, with violation below 5% on both windows and lower energy than pure BB on the peak window.

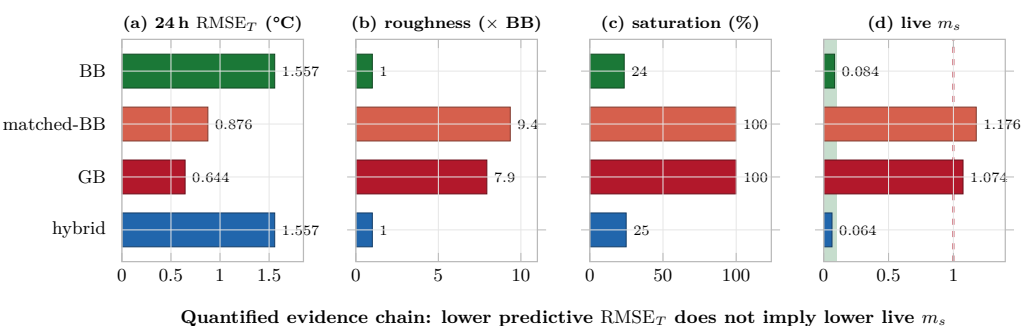


Figure 4. The fidelity–utility paradox at a glance. For the four thermostatic-PPO backends the measured chain runs (A) 24 h rollout RMSE → (B) action-surface roughness → (C) policy saturation → (D) live maintenance score m_s . The higher-fidelity backends (calibrated GB, matched-resolution BB) win on (A) yet are rougher, saturate, and collapse in (D) ($m_s > 1$), while the coarse hourly BB and the hybrid stay usable, and the contradiction between (A) and (D) is the paradox. For the hybrid, panel (A) shows the BB rollout RMSE because the hybrid rolls out on BB; GB enters only as a frozen reward censor, so the hybrid has no independent 24 h predictive RMSE (hence the “—” in Table 4). Panel (D) reports the cross-window mean of the peak and typical maintenance scores (per-window values in Table 4).

Table 4. Central fidelity–utility paradox. The four thermostatic-PPO training backends of the taxonomy (Table 1), each trained on its surrogate and evaluated zero-shot in closed loop on the live BOPTTEST `bestest_air` runtime over the peak/typical 14-day windows. Predictive fidelity (24 h rollout RMSE; lower is more accurate) is set against live control utility (maintenance score m_s , with $m_s > 1$ unusable, and comfort violation). Higher predictive fidelity does not imply usability: both finer backends collapse, whereas the role-separated hybrid is robust. The matched-resolution BB row isolates the model-class confound (Section 4.3); per-window live closed-loop RMSE and energy are broken out in Supplementary Figure S7.

Training backend	Architecture	Step (s)	24 h RMSE (°C)	m_s (pk/typ)	Viol. % (pk/typ)	Verdict
BB hourly	black-box	3600	1.557	0.073 / 0.095	1.5 / 4.4	usable
matched BB	black-box	900	0.876	1.142 / 1.211	85.6 / 91.4	collapse
calibrated GB	grey-box RC	900	0.644	1.046 / 1.102	77.1 / 82.4	collapse
hybrid	BB + GB censor	900	—	0.087 / 0.041	4.7 / 2.4	robust

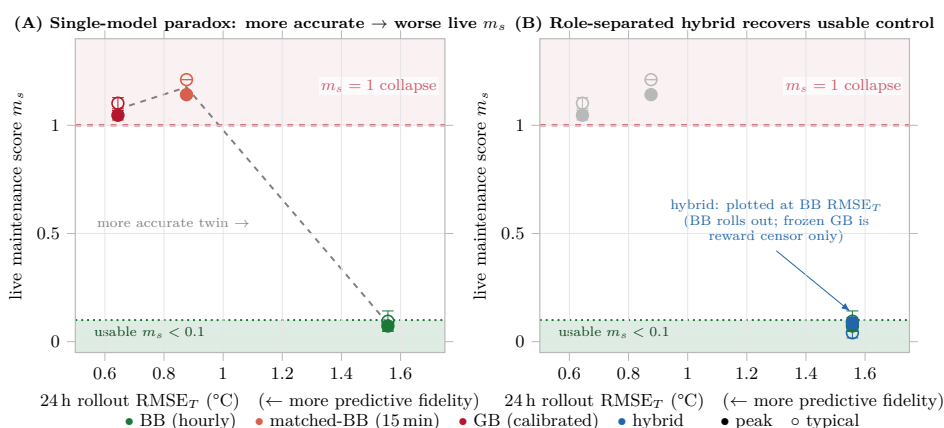


Figure 5. Predictive fidelity versus live control utility. Lower rollout RMSE can raise live m_s ; the hybrid returns to the usable region. Filled/open markers denote peak/typical windows; markers are the canonical seed-42 m_s and error bars are ± 1 seed s.d. over $N = 3$ seeds for all four backends, including direct GB (Table S14).

The decomposition $m_s = r_{\text{time}} + r_{\text{sev}}$ (Eq. 3) exposes the failure structure directly (Figure 6). Direct GB's canonical (seed-42) peak $m_s = 1.046$ ($N = 3$ mean 1.071 ± 0.022 ; the typical window is seed-42 1.102 , $N = 3$ mean 1.131 ± 0.026 ; both seed-stable, Table S14) is dominated by $r_{\text{time}} \approx 0.77$ (out of band 77% of the time) plus a large worst-case severity $r_{\text{sev}} \approx 0.28$; the hybrid's peak $m_s = 0.087$ is almost entirely a small r_{sev} with $r_{\text{time}} < 0.05$. The hybrid therefore does not merely lower the average error; it removes the sustained band departures that dominate the direct-GB score.

Taken alone, the main-text evidence already settles the central paradox: Table 4 shows that both finer backends (the matched-resolution BB and the more accurate direct GB) are unusable in live control despite their higher predictive fidelity, and Figure 5 shows that lower rollout RMSE does not imply lower live m_s ; the matched-resolution BB row in the same table closes the model-class confound. The Supplementary tables add only seed-level and artefact-level audit behind these numbers.

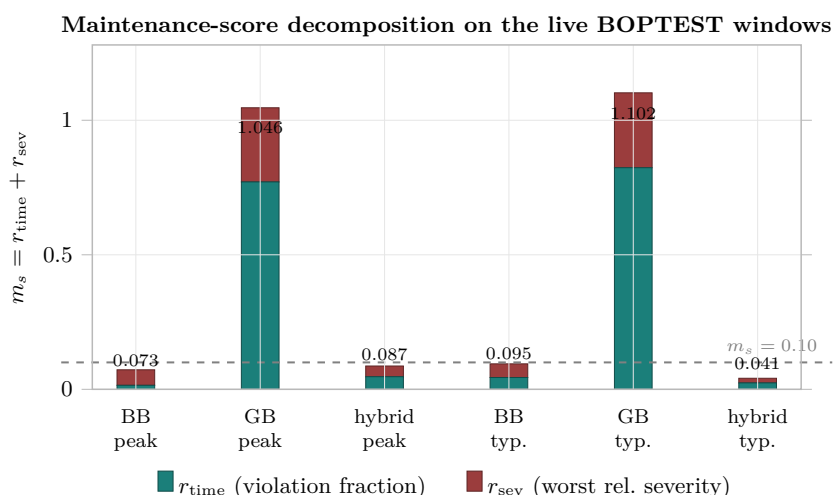


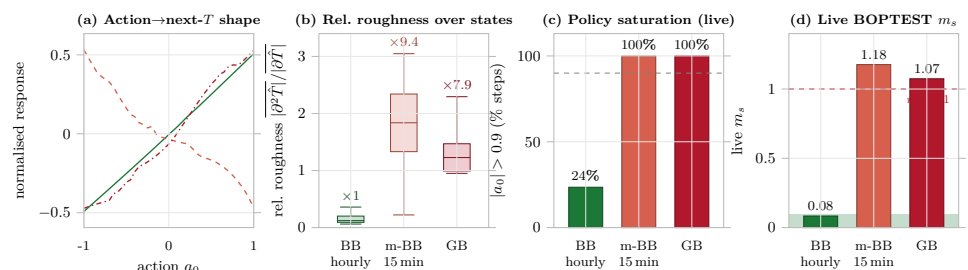
Figure 6. Live BOPTEST maintenance-score decomposition. Direct GB fails through sustained out-of-band time; pure BB and the hybrid keep duration and severity small.

4.3. Mechanism: temporal coarseness, roughness, and saturation

The high-fidelity backends collapse because their action-response surfaces are rougher, which drives near bang-bang policy saturation and live failure, whereas the role-separated hybrid

avoids this by separating smooth rollout from GB-based model-disagreement censoring. The mechanism is the measured chain from surrogate response geometry to policy behaviour to live BOPTEST outcome. This subsection is not an additional controller comparison; it explains why the single-model backends collapse and why the hybrid remains usable.

Figures 7 and S9 explain the collapse: direct GB learns a bang-bang law that places extreme actions in temperature-error regimes the live building does not support. The full training-trajectory loss-landscape is not measured, but the surrogate-side roughness and policy-side saturation that drive it *are* (Fig. 7); which micro-cause dominates is deferred to future work. Figure 7 is itself sufficient to read the measured mechanism: both collapsing single-model backends are rougher ($9.4\times$ and $7.9\times$ the usable BB), saturate the action (100% vs. 24% of steps), and fail live BOPTEST ($m_s > 1$). The time-domain signature is direct GB saturating and leaving the comfort band while the coarse BB and the hybrid stay in band (Figure 8); the Supplementary table records the numerical audit behind these plotted values.



Measured mechanism: rougher action surface $\rightarrow$ policy saturation $\rightarrow$ live collapse (association, not a proven causal law)

Figure 7. Measured roughness mechanism. The collapsing single-model backends expose rougher action-response surfaces, produce near bang-bang saturation, and fail live BOPTEST; the pattern is mechanistically consistent with the fidelity–utility paradox but does not by itself prove the full PPO loss-landscape causality.

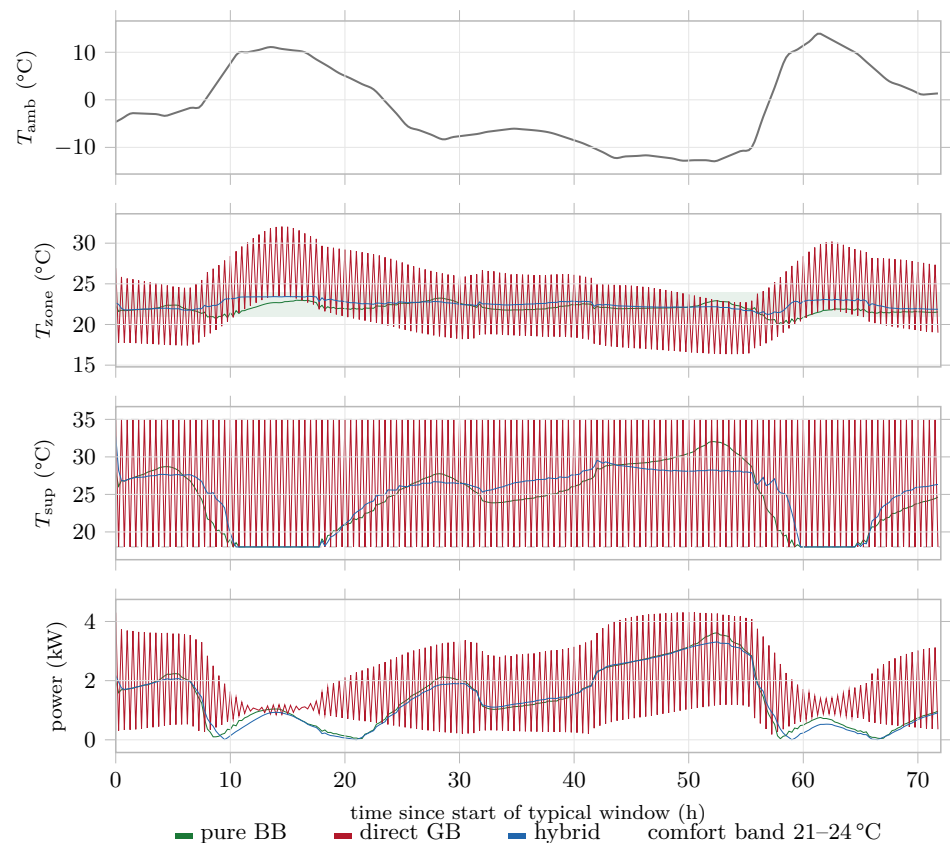


Figure 8. Live closed-loop behaviour (typical window). Zone temperature against the comfort band, with ambient disturbance and HVAC power, for the pure-BB, direct-GB, and hybrid controllers on the live BOPTEST runtime: direct GB saturates the actuator and leaves the comfort band, whereas the coarse BB and the hybrid track it and stay in band.

The direct-GB failure could be dismissed as a grey-box model-class artefact; a matched-config control closes this confound. Retraining the *same black-box* BB on the matched 15-minute corpus makes it a strictly *more accurate* predictor (24 h RMSE 0.876 vs 1.557 °C) yet, trained under the identical pure-BB recipe, it *collapses* in closed loop ($m_s = 1.142/1.211$, 85.6/91.4% violation; matched-BB row of Table 4), on a par with direct GB and far worse than the hourly-trained controller.

This ablation isolates the model-class confound: improving BB's temporal resolution and predictive RMSE does not improve downstream utility; it destroys it, so BB's useful property is not its black-box class alone. Two mechanisms are entangled in this coarse-versus-fine contrast: temporal coarse-graining yields an optimization-friendly (smoother) response surface, but the hourly BB is also deployed with its hourly increment applied unchanged at the 900 s step (four times the dimensionally-consistent per-step change), so part of both its smoothness and its 1.557 °C error is a train/deploy timestep mismatch rather than corpus resolution alone. A Δt -rescaled control settles which part is load-bearing: keeping the same hourly BB but rescaling its increment to the 900 s runtime step collapses the controller on every seed ($m_s = 1.19/1.22$, $N = 3$; Table S14), placing it with the matched-resolution and direct-GB controls. The hourly BB's usability therefore rests on its large, temporally-coarse per-step increment, not on the corpus alone; an intermediate-resolution sweep (e.g. 1800 s) remains future work. The outcome across the three single-model backends is therefore not a monotonic fidelity–utility curve but a two-level contrast: the coarse hourly BB is usable, while both finer backends collapse (matched-resolution BB $m_s = 1.14/1.21$ and the still more accurate direct GB $m_s = 1.046$ are both unusable, and

the ordering inside the collapse region is non-monotonic in accuracy). This reframes the hybrid as an explicit separation of smoothing (BB rollout) from fidelity (frozen GB censor): PPO needs not only accurate predictions but an optimization-friendly training landscape.

Retrained on three fixed seeds ($\{42,43,44\}$), each backend keeps its regime: pure BB and the hybrid hold every seed below the 5% violation bar on both windows, while the matched-resolution BB collapses on every seed ($m_s = 1.14/1.21$, $\text{std} < 0.001$) and the direct GB collapses on every seed ($m_s = 1.07 \pm 0.02 / 1.13 \pm 0.03$ peak/typical), so the usable-versus-collapse verdicts are seed-stable (Supplementary Table S14, Fig. S9); the single-seed peak edge of pure BB over the hybrid does not persist across seeds (HDRL single-seed; MORL over $N=5$).

The smoothing is directly measurable on the surrogates, independent of any controller: sweeping each backend's one-step map $\hat{T}(a_0)$ over a grid of states, the scale-free *relative roughness* (mean curvature $\div$ mean slope) orders cleanly with usability (Table S15): the usable hourly BB is smoothest (0.17), while both collapsing backends are several times rougher ($9.4\times$ matched-resolution BB, $7.9\times$ GB). Relative roughness is necessary but not sufficient, however: a constant Δt -rescaling preserves the curvature/slope ratio, so the Δt -rescaled hourly BB has the *same* relative roughness (0.17) as the usable hourly BB yet collapses (Table S14). The operative surrogate property is therefore the coarse per-step increment magnitude (temporal coarseness), of which relative roughness measures one facet, rather than scale-free smoothness alone. This is the measured surrogate-side counterpart of the policy-side near-bang-bang action gap (Figure 7 and Fig. S8); the raw slope/range are deliberately not used as the metric, being confounded by the step length.

4.4. Hybrid robustness and controller-family boundary tests

The canonical hybrid operating point was selected by a thermostatic sweep over the temperature-disagreement weight $\lambda_T \in \{0.05, 0.10, 0.15\}$ at fixed $\lambda_P = 5 \times 10^{-5}$ (Table S16). The mid setting $\lambda_T = 0.10$ is canonical: it retains the energy advantage over pure BB while avoiding the stronger comfort degradation seen at the weaker (0.05) and stronger (0.15) censor settings. Only the canonical point's live-BOPTEST KPIs are retained as a frozen artifact; the bracketing points ($\lambda_T = 0.05$ and 0.15) served selection only and were not carried to the full live two-window protocol, which is why the controller-family λ_T comparison legitimately shows thermostatic PPO at its two live-evaluated points, $\lambda_T = 0$ (pure BB) and $\lambda_T = 0.10$ (hybrid).

A second negative control tests whether GB helps as a policy *initializer* rather than a reward censor: a thermostatic policy pretrained on GB and warm-start-finetuned on live BOPTEST is compared against an otherwise identical scratch finetune (Table S17). It does not help; warm-starting raises m_s by two to three times on both windows: GB belongs in the reward as a censor, not in the weights as an initializer.

Transfer-gap diagnostics.

Transfer-gap diagnostics confirm the same ordering from the surrogate side: direct GB shows by far the largest surrogate-to-live mismatch (a learned bang-bang exploit whose policy sits at the opposite action bound from BOPTEST throughout, $g_a \approx 2.0$), while the hybrid shows the smallest and stays BOPTEST-consistent longest. The per-controller diagnostics (temperature gap, surrogate-vs-live Δm_s , action-gap norm, and first divergence step) and their protocol are reported in Table S18.

HDRL, MORL, and the PI reference.

The physics-censor recipe is controller-family specific. The preceding sections establish the paradox and its hybrid remedy within the thermostatic PPO family; two further families test whether the recipe transfers. These are boundary tests, not additional proof of the

paradox, so their detailed sweeps, architectures, pipelines, and seed analyses are kept in the Supplementary Material.

HDRL: censor-weight specificity.

A seasonal hierarchy routes control to winter/summer PPO setpoint specialists that share the 17D observation and supply-temperature action (Figure S12). The thermostatic-optimal censor weight $\lambda_T = 0.10$ does *not* transfer: because each specialist is already comfort-aware, the GB temperature censor over-regularizes the low-level loop and m_s degrades monotonically with λ_T (peak violation roughly quadruples, 6.1% $\rightarrow$ 23.0%). HDRL is best at $\lambda_T = 0$, keeping only the power channel ($\lambda_P = 5 \times 10^{-5}$); the correct censor strength is therefore family-specific, not a universal weight (Table S19, Fig. S13).

MORL: observation geometry and seed stability.

A four-stage MORL pipeline (surrogate pretrain $\rightarrow$ preference adaptation $\rightarrow$ live-BOPTEST finetune $\rightarrow$ yearly evaluation) on the power-only hybrid backend is viable only once the observation is widened from 5D to 17D: the current-code 5D interface fails ($\text{RMSE}_T = 2.721^\circ\text{C}$, 37.8% violation, $m_s = 0.680$) while the 17D forecast-augmented vector recovers a usable policy ($\text{RMSE}_T = 0.720^\circ\text{C}$, 4.9%, $m_s = 0.099$). The bottleneck is observation geometry, not the reward scalarization, consistent with forecast-augmented RL-HVAC control [4]. MORL yields a usable comfort-energy Pareto front, but an $N = 5$ seed extension turns the canonical operating points into wide distributions (CV 0.42–0.61) that fail a pre-specified stability test, promising in mean but not deployment-stable without explicit stabilization (the 5D $\rightarrow$ 17D observation change is shown schematically in Figure S15 and quantified in Figure S16; the $N = 5$ Pareto seed analysis is retained in the Supplementary Material, Tables S20–S21).

Synthesis against the PI reference.

The BOPTEST built-in PI controller is a reproducible reference, comfort-poor but energy-frugal (12-month means: $\text{RMSE}_T = 3.40^\circ\text{C}$, 63.6% violation, $m_s = 0.910$). All evaluated RL/MORL agents reach m_s well below PI, dominating it on comfort at comparable or lower energy. The cross-family maintenance-score decomposition (Supplementary Table S29) makes the pattern explicit: the failures (direct GB, PI) carry a large time-in-violation term $r_{\text{time}} \gtrsim 0.6$ (PI by chronic under-heating, direct GB by saturated overshoot), whereas the usable controllers (pure BB, hybrid, MORL 17D) keep both r_{time} and r_{sev} small.

The controller-learning experiments establish four claims. (i) Predictive fidelity $\neq$ training utility: direct GB is the more accurate twin yet fails as a rollout environment ($> 77\%$ violation), and the matched-resolution ablation (matched-BB row of Table 4) shows the *same* black-box BB also collapses once its resolution (and accuracy) rises, isolating the fidelity/smoothing trade-off, not the model class. (ii) Role separation works: BB supplies smooth rollout dynamics and frozen GB a disagreement censor, giving the hybrid ($m_s = 0.087/0.041$). (iii) The censor strength is controller-family specific (HDRL best at $\lambda_T = 0$). (iv) MORL is viable only with the 17D interface, with high $N = 5$ seed variance. The fixed `bestest_air` recipe is then transferred to the hydronic BOPTEST family.

4.5. Transfer boundary: calibration transfers, frozen policy does not

4.5.1. Protocol, testcases, and metrics

Transfer is component-level: the inverse-calibration pipeline ports across the hydronic family (rollout RMSE reduced 60.2–87.8%; C_{zon} re-identified at a near-uniform $1.918 \pm 0.032\times$), whereas the frozen policy transfers only in some regimes. The predictive-validation and central-control experiments established the source-case recipe on BOPTEST `bestest_air`: a BB rollout surrogate supplies smooth control-oriented dynamics, a cali-

brated GB RC–NeuralODE supplies a frozen physics-informed model-disagreement censor, and the thermostatic PPO policy trained on the resulting hybrid backend is the strongest verified controller on the targeted windows. This transferability experiment does not test universal controller generalization. It decomposes transfer into two components: frozen-policy transfer and surrogate-recalibration transfer. This distinction prevents a misleading binary conclusion.

The claim is deliberately bounded: the transferability experiment evaluates transfer to three pre-selected single-zone hydronic cases under documented actuator adapters and pre-specified recalibration, with target-testcase controller fine-tuning explicitly excluded, separating *controller* transfer from *surrogate-recalibration* transfer so the evidence is reported component-wise.

The transferability hypotheses, thresholds, and adapters were pre-specified before any non-bestest_air run, so every number below was predictable but not predicted; full reproduction details are in the Supplementary Material.

Testcases, actuator adapters, and recalibration regimes.

The three target testcases span an increasing difficulty ladder relative to *bestest_air*; all are single-zone or single-zone-like, heating-capable, and structurally non-identical to the source case.

A pre-specified per-testcase adapter $\mathcal{A}_k$ maps the frozen policy’s supply-temperature command (Eq. (2)) to each target’s BOPTEST override inputs, committed and smoke-tested for directional validity before any run; the recalibration regimes keep the controller frozen and vary only how much of the surrogate adapts (adapter definitions and regimes in Supplementary Tables S23–S24, Figure S20).

The three targets differ in heat source, distribution path, and actuator set, but share the same single-zone energy balance (Figure S21). For a hydronic zone the governing first-order balance is

$$C_{\text{zon}} \frac{dT_{\text{zon}}}{dt} = \dot{Q}_{\text{hyd}}(u_t) + \frac{T_{\text{amb}} - T_{\text{zon}}}{R_{\text{env}}} + \dot{Q}_{\text{gain}}, \quad (6)$$

where $\dot{Q}_{\text{hyd}}$ is the delivered hydronic heat (whose actuation path, heat pump, boiler/radiator, or district valve, changes per testcase), R_{env} the envelope resistance, and $\dot{Q}_{\text{gain}}$ the internal/solar gains. Transfer changes the *delivery path* $\dot{Q}_{\text{hyd}}$ and the actuator interface, but C_{zon} is the shared lumped physical coordinate that Stage B re-identifies, which is why its re-identified value transfers (Section 4.5.3).

Table S25 summarizes the identified physics. The equivalent thermal mass $C_{\text{zon}}/c_{p,\text{air}}$ is roughly $2\times$ the *bestest_air* value for every hydronic case (added water-loop and surface mass). Mean delivered power exposes the regime: approximately 0.7–2.8 kW for the residential cases (conduction/slow-hydronic dominated) but 76.9 kW for the commercial case (ventilation/AHU-dominated), about $28\times$ the heat-pump target, the physical reason the commercial controller passes comfort yet overshoots energy by +35.3%.

The controller-side pass threshold is normalized by each testcase’s built-in PI baseline ($\tau = 1.25 m_{s,\text{PI}}$, with the 1.25 value pre-specified rather than chosen post hoc); energy is reported on a separate axis as the percentage change against PI; surrogate-side transfer is the relative rollout-RMSE improvement after full Stage A/B/C recalibration; and the physical transfer diagnostic is the re-identified capacitance ratio ρ_C against the source value $C_{\text{air}} = 4.413 \times 10^5 \text{ J/K}$ (all column formulas in the caption of Table 5). Two-axis reporting (m_s threshold and ΔE) is necessary because the commercial testcase passes the m_s bound while consuming substantially more energy than PI.

4.5.2. Controller-side transfer verdicts

Table 5 contains two stories. The surrogate component transfers strongly: full Stage A/B/C improves RMSE by 60.2–87.8% on all three targets. The frozen controller does not transfer uniformly: both residential cases fail the $1.25\times$ -PI threshold, while the commercial stretch passes it at a +35.3% energy penalty. This component-level boundary (identified physics transfers, the frozen control law does not) is consistent with building heterogeneity requiring local adaptation rather than zero-shot reuse [46,47].

Table 5. Transferability matrix. Columns: $m_{s,RL}$ and $m_{s,PI}$, the maintenance scores of the RL controller and the PI baseline; pass threshold $\tau = 1.25\,m_{s,PI}$; the controller verdict is PASS when $m_{s,RL} \leq \tau$; energy change $\Delta E = 100\,(E_{RL} - E_{PI})/E_{PI}$ (%); raw and full-recalibration rollout RMSE with relative gain $G_{RMSE} = 100\,(RMSE_{raw} - RMSE_{full})/RMSE_{raw}$ (%); capacitance ratio $\rho_C = C_{zon}/C_{air}$ against the source $C_{air} = 4.413 \times 10^5$ J/K. [†] denotes threshold PASS with energy penalty.

Testcase	$m_{s,RL}$	$m_{s,PI}$	τ	Ctrl.	$\Delta E\%$	Raw RMSE	Full RMSE	Gain%	ρ_C
hydronic heat pump	0.665	0.464	0.579	FAIL	-7.3	1.421	0.565	60.2	1.892
hydronic (boiler)	0.976	0.750	0.938	FAIL	-5.8	2.666	0.335	87.4	1.954
commercial hydronic	0.431	0.628	0.785	PASS [†]	35.3	1.952	0.238	87.8	1.909

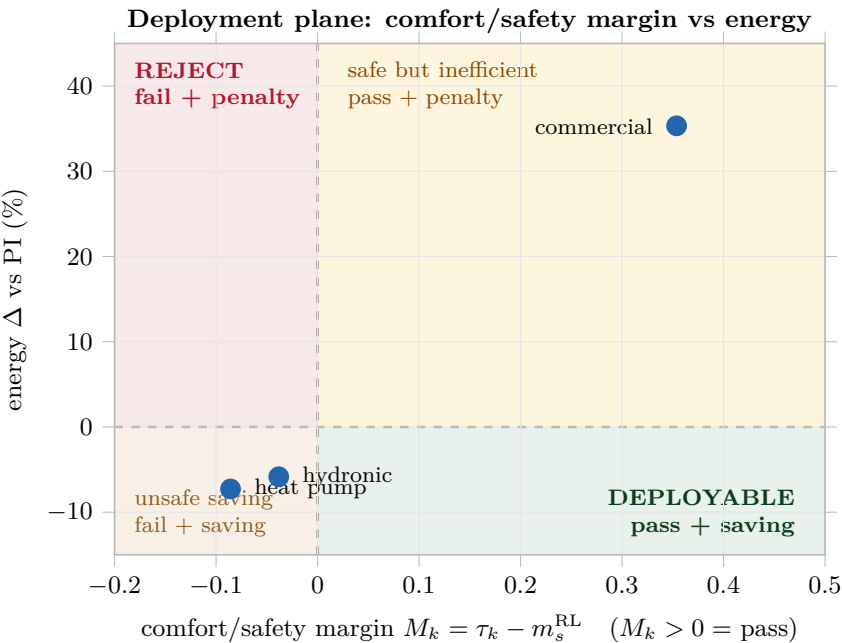


Figure 9. Comfort–energy deployment plane. Residential hydronic cases save energy but fail the comfort threshold; commercial hydronic passes the threshold with a 35.3% energy penalty. No frozen-policy transfer is deployment-ready.

The three testcases confirm the component-level split (Table 5, Figure 9). Both residential cases fail the $1.25\times$ -PI controller threshold while *saving* energy (heat-pump $m_s = 0.665$ at -7.3% ; boiler $m_s = 0.976$ at -5.8%), yet full Stage A/B/C recalibration succeeds on both surrogates (RMSE_T to 0.565 and 0.335 °C; 60.2/87.4% gains; $C_{zon} = 1.892/1.954\times$). The pre-specified commercial “stretch” case is mixed: the frozen controller *passes* the m_s threshold (0.431 vs PI 0.628) but only at +35.3% energy (threshold-pass, not deployment-ready), while its surrogate side is again strong (87.8% RMSE gain, $C_{zon} = 1.909\times$, contradicting the version-locked pre-specified scale-dependent $[3, 10]\times$ prediction). Across all three, surrogate recalibration transfers; the frozen controller does not transfer uniformly.

4.5.3. Surrogate-side transfer

The surrogate component transfers on all three targets: full Stage A/B/C cuts rollout RMSE by 60.2–87.8%, and the re-identified zone capacitance clusters at $\rho_C = 1.918 \pm 0.032 \times$ the source, read as an *effective* single-zone RC signature rather than a physical invariant, a curious $N = 3$ regularity, not a law. The full C_{zon} -consistency analysis, the surrogate-side recalibration curves, and the pre-specified hypothesis closure and predictions are reported in the Supplementary Material (Figure S28; Tables S25, S27, S34, and S28).

5. Discussion

The experimental sequence closes the paper's four claims without treating every controller family as part of the primary paradox. First, calibrated GB is the best predictor, but direct use of GB and matched-resolution BB as PPO rollout environments falsifies the assumption that predictive fidelity alone implies training utility. Second, the hybrid backend supports the role-separation hypothesis: BB remains the smooth rollout substrate and frozen GB supplies only a reward-shaping disagreement censor. Third, HDRL and MORL show that this regularization is controller-family specific and observation-dependent. Fourth, hydronic transfer is component-level: inverse calibration transfers more reliably than the frozen policy.

Predictive fidelity and training utility are distinct, and over part of the range opposed. The calibrated twin GB is the far better predictor (24 h rollout RMSE 0.644 vs 1.557 °C) yet, used directly as the training environment, yields an unusable controller (seed-42 $m_s = 1.046$; $N = 3$ mean 1.071, seed-stable collapse; $> 77\%$ violation), while the weaker BB yields a usable one ($m_s = 0.073/0.095$). A matched-resolution closed-loop ablation settles the obvious confound: the *same* black-box BB retrained at 15 minutes is strictly *more accurate* (0.876 °C) yet also collapses ($m_s = 1.14/1.21$; Table 4), so the operative variable is the fidelity/smoothing trade-off induced by temporal resolution, not the model class, and a matched-corpus decomposition splits the fidelity gain into a corpus-resolution term and a Stage A/B/C calibration term whose relative size is decomposition-path-dependent, so it is reported symmetrically rather than as a single number (25.4% calibration under the additive matched-BB split versus 90.1% under the raw-GB-backbone split; Section 4.1). This is interpreted as an instance of model exploitation in model-based RL, a distribution-shift effect [10,41]: the sharper response surface of the high-fidelity backends (7.9–9.4× rougher; Supplementary Table S15) is exploited by policy-gradient search into a near bang-bang law that does not survive transfer. A Δt -rescaling control isolates the operative property: a backend with the *same* scale-free roughness but a fine per-step increment also collapses (Table S14), so what matters is the coarse increment magnitude, not smoothness alone.

Role separation, not ensembling, is what makes the hybrid work. Separating the two roles (BB for smooth dynamics, a frozen GB as a reward-shaping censor) recovers the best cross-window robustness ($m_s = 0.087/0.041$, sub-5% violation on both windows) at an $85\times$ training speed-up. The hybrid should therefore not be read as a generic ensemble of two predictors: if it were an accuracy-oriented ensemble, GB would naturally dominate the rollout. The experiments show the opposite. The physically richer model is useful only after its role is restricted to a detached model-disagreement censor, while the smoother BB remains the training substrate.

The physics-regularisation recipe is controller-family specific, and transfer resolves into a component-level boundary. The optimal censor weight is $\lambda_{\text{temp}} = 0.10$ for thermostatic PPO but 0 for the hierarchical agent, which already encodes enough forecast/structural context that an external censor over-constrains it [11,12], so a single “best” regularisation strength cannot be reported without naming its controller family. This falsification of λ -universality rests on the two independently swept families (PPO versus

HDRL); the 17D MORL agent adopts $\lambda_{\text{temp}} = 0$ by analogy with the HDRL sweep and is not counted as independent evidence for the claim. For transfer, the calibration *pipeline* ports (rollout RMSE down 60.2–87.8% on the hydronic family; C_{zon} re-identifies at a near-uniform $1.918 \pm 0.032 \times$ ratio across an order-of-magnitude volume change (a family-level regularity across three cases, not a claimed physical invariant)), whereas the frozen *policy* does not transfer uniformly: the commercial case passes the $1.25 \times$ -PI comfort threshold but at a 35.3% energy penalty, while the residential cases fail it zero-shot. The contribution is thus less a new controller than a corrected, quantified account of *why* surrogate-trained controllers succeed or fail, under a pre-specified protocol that lets H1/H3 be stated as *falsified* rather than quietly dropped.

Limitations.

Evaluation is simulation-only on a single-zone `bestest_air` emulator, with multi-zone and richer topologies untested and transfer limited to a hydronic family. Several caveats bound the central claim. First, the fidelity–utility relationship is established at three discrete operating points (hourly BB, matched 15-min BB, and the calibrated GB) rather than as a continuous fidelity–utility curve: the seed replication ($N = 3$, seeds $\{42, 43, 44\}$) shows that the usable controllers, the matched-resolution collapse, and the direct-GB collapse are all seed-stable (Table S14), but the fidelity axis is still sampled at three discrete operating points rather than as a distribution, and an intermediate-resolution sweep (e.g. 1800 s) is left to future work. Second, the only comparison baseline is BOPTEST’s built-in PI, a relatively weak reference, so controller utility is established comparatively against a modest baseline rather than against a strong tuned rule-based, Guideline-36, or MPC controller on the same RC model. Third, the surrogate power channel is reproduced less accurately than temperature (calibrated power-head MAE 482 W, about 9% of $P_{\text{max}} = 5500$ W, rising to ~ 2.3 kW in the partial-calibration diagnostic regime); the reported energy KPIs (Supplementary Figure S7) and the transfer ΔE (Table 5, e.g. the commercial +35.3%) are live-versus-live BOPTEST measurements and do not inherit this surrogate error, but the weaker power channel limits predictive-fidelity claims about the power head and adds noise to the training-time energy reward, which uses the surrogate power prediction, and a cumulative surrogate-versus-BOPTEST energy balance for at least one window is left to future work. Fourth, the controller-family specificity of the censor weight that falsifies H3 rests on a non-uniform sweep density (a dense four-point λ_{temp} sweep for HDRL but only two measured endpoints for thermostatic PPO and a single adopted configuration for MORL), so it is read as consistent with family-specific regularisation rather than as a fully-swept per-family optimum. Finally, the paradox mechanism is *measured* statically on each surrogate’s one-step action map and corroborated by the matched-resolution ablation, but a loss-landscape analysis along the PPO trajectory and generality to off-policy algorithms (SAC [55], TD3) remain open; statistical support otherwise varies by family (MORL $N = 5$, neutral $m_s = 0.187 \pm 0.078$ with CI far below the PI 0.910; HDRL single-seed), and the backend $\times$ observation grid is only partially crossed.

6. Conclusion

This study asked whether a surrogate with higher predictive fidelity is automatically a better environment in which to train a reinforcement-learning HVAC controller, and tested the question across three controller families and a hydronic family of transfer testcases under a pre-specified protocol. The answer is no. Only the temporally coarse, hourly-trained black-box surrogate (BB) trains a usable controller: the most accurate physical twin (GB; 0.644 °C 24-hour rollout RMSE) collapses on every seed (seed-42 $m_s = 1.046$; $N = 3$ mean 1.071, seed-stable; comfort violation above 77%), and the *same* black-box architecture

retrained at the finer 900 s resolution becomes a *more* accurate predictor (0.876 °C) yet also fails as a training environment ($m_s = 1.14/1.21$, violation above 85%), a fidelity–utility paradox (H1 falsified) whose operative variable is the surrogate’s coarse per-step increment magnitude (temporal resolution), not the surrogate model class; a matched-corpus attribution splits the predictive gain into corpus-resolution and calibration terms whose relative size is decomposition-path-dependent (25.4% vs 90.1% calibration across the two splits) and is reported symmetrically, not as a single number. Separating the two roles resolves it: using the black-box surrogate for smooth rollout dynamics and a frozen physical twin as a reward-shaping censor (H2 supported) recovers the best cross-window robustness of the study ($m_s = 0.041$ typical, violation below 5%) at an $85\times$ training speed-up. The censor strength is controller-family specific ($\lambda_{\text{temp}} = 0.10$ for thermostatic PPO versus 0 for the hierarchical agent; H3 falsified by these two independently swept families, the MORL setting adopted by analogy), and transferability resolves into a component-level boundary (H4): the inverse-calibration pipeline transfers; rollout RMSE falls by 60.2–87.8% and the zone capacitance re-identifies at a near-uniform ratio ($1.918 \pm 0.032\times$ the source), whereas the frozen control policy does not transfer zero-shot. The contribution is therefore less a new controller than a corrected, quantified account of why surrogate-trained controllers succeed or fail, together with a reusable role-separating recipe.

Several future directions follow directly from the limitations of Section 5: extending the study from a single zone to multi-zone and mixed-topology buildings; extending the now-measured static response-surface roughness (Supplementary Table S15) to a formal loss-landscape analysis along the full PPO training trajectory; improving the heating-power channel of the surrogate so it is a validated rather than an auxiliary output; broadening transfer beyond the hydronic family and pairing the transferable calibration pipeline with light policy adaptation instead of zero-shot reuse; and, ultimately, closing the remaining sim-to-real gap through deployment on a physical building.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/ai1010000/s1>: the complete nomenclature, the experimental protocol, the surrogate-calibration and controller-training details, the BOPTEST run-time-environment control-loop description, and supporting summaries for the predictive-validation, controller-utility, and transfer-boundary evidence. The Supplementary Material is a technical support layer; every quantity needed to interpret the scientific claims is reported in the main text.

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